1 The Chain Rule

Consider the path $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by

$$\begin{aligned}\mathbf{r}(t) &= (x(t), y(t), z(t)) \\ &= (\sin(t), \cos(t), 2t)\end{aligned}$$

and the multivariable function $f : \mathbb{R}^3 \to \mathbb{R}$ defined by

$$f(x,y,z) = x^2 + y^2 + z^2$$

1. Since the range of $\mathbf{r}$ is the domain of f it makes sense to form the composition $f \circ \mathbf{r}$. Write out

   $$(f \circ \mathbf{r})(t) := f(\mathbf{r}(t)) =$$

   Fill in the domain and codomain of $f \circ \mathbf{r}$ below

   $$f \circ \mathbf{r} : \qquad \to$$

2. Calculate the following:

   $$\begin{aligned}\nabla f(x,y,z) &= \\ \mathbf{r}'(t) &=\end{aligned}$$

3. Now calculate without simplifying

   $$(f \circ \mathbf{r})'(t) =$$

4. Compare your answer to 3 with your answers to 2. Make a conjecture about how $(f \circ \mathbf{r})'(t)$ is built out of $\nabla f(x,y,z)$ and $\mathbf{r}'(t)$. Does this look similar to the single-variable Chain Rule?

Proposition 1.1 (Chain Rule)**.** *If $\mathbf{r} : I \subseteq \mathbb{R} \to \mathbb{R}^3$ is differentiable at $t \in I$, and $f : \mathbf{r}(I) \subseteq D \subseteq \mathbb{R}^3 \to \mathbb{R}$ is continuously differentiable at $\mathbf{a} = \mathbf{r}(t) = (x(t), y(t), z(t))$, then $f \circ \mathbf{r} : I \to \mathbb{R}$ is differentiable at t, and*

$$(f \circ \mathbf{r})'(t) = \nabla f(x(t), y(t), z(t)) \cdot \mathbf{r}'(t).$$

If we write $F = f \circ \mathbf{r}$, then in Leibniz notation the Chain Rule becomes

$$\frac{dF}{dt} = \frac{\partial F}{\partial x}\frac{dx}{dt} + \frac{\partial F}{\partial y}\frac{dy}{dt} + \frac{\partial F}{\partial z}\frac{dz}{dt}.$$

Before proving Proposition 1.1, let's work an example.

5. If $f(x,y,z) = x^2 - y^3 + xyz$, and $x = 6t + 7, y = \sin(2t), z = t^2$, verify the Chain Rule by finding $\frac{df}{dt}$ in two different ways: by differentiating the composition, and by using the Chain Rule. Which way is easier?

2 Proof of the Chain Rule

Name:_____

3 Exercises

1. If $z = x^2y + 3xy^4$, where $x = \sin(2t)$ and $y = \cos(t)$, find $\frac{dz}{dt}$ when $t = 0$.

2. The pressure P (in kPa), volume V (in L), and temperature T (in K) of one mole of an ideal gas are related by the equation $PV = 8.31T$. Find the rate at which the pressure is changing when the temperature is 300 K and increasing at a rate of 0.1 K/s and the volume is 100 L and increasing at a rate of 0.2 L/s.

3. For each function below, use the Chain Rule to find $\frac{dz}{dt}$.

   (a) $z = x^2 + y^2 + xy$, $x = \sin(t)$, $y = e^t$

(b) $z = \cos(x+4y)$, $x = 5t^4$, $y = \frac{1}{t}$

(c) $z = \sqrt{1+x^2+y^2}$, $x = \ln(t)$, $y = \cos(t)$

(d) $z = \tan^{-1}(y/x)$, $x = e^t$, $y = 1 - e^{-t}$

4. For each function below, use the Chain Rule to find $\frac{dw}{dt}$.

   (a) $w = xe^{y/z}$, $x = t^2$, $y = 1 - t$, $z = 1 + 2t$

   (b) $w = \ln(\sqrt{x^2 + y^2 + z^2})$, $x = \sin(t)$, $y = \cos(t)$, $z = \tan(t)$